

PV–PORT 180 A port board

Rev F0 | 2026–10–04 | Schematics authored by Chinmoy Bhuyan | Ports A (PV) and B (bus/battery) of PV–P100/110, common negative BUS–

01 Connectors and sensor supply

PORT 26–way to CTRL–C2000, 14–way to SYS–IO–AUX, ID + hold readback, +24V → +3V3S buck, AGND tie, PE bond

File: sheets/01_io.kicad_sch

02 Port A power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/02_portA_power.kicad_sch

03 Port A sensing and hold logic

VA bus side, VAX terminal side (AMC3330), IA current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/03_portA_sense.kicad_sch

04 Port B power path

Terminals, fuses, SPD, EMI filter, contactor, precharge relay, coil drives, hold stage, shunt, discharge + time limit, bleeder

File: sheets/04_portB_power.kicad_sch

05 Port B sensing and hold logic

VB bus side, VBX terminal side (AMC3330), IB current (AMC3302), two–channel hold logic, proof–test injection, readback RB1/RB2

File: sheets/05_portB_sense.kicad_sch

06 Insulation measurement

AUX1 = V(PE–BUS–), AUX2 = V(PE–A+) (AMC3330); 992 k test strings A+ → PE, BUS– → PE, each with a common–source SiC pair (TPSI3050)

File: sheets/06_imd.kicad_sch

Common negative: port A and port B share BUS– (non–isolated buck–boost), so both ports are one HV domain; the main contactors switch the + pole only, fuses are in both poles. Chassis–mounted parts (terminals, fuses, CM choke, contactors, RST 200 resistors, thermostats, shunt) are drawn here with their real MPN or an RFQ / CUSTOM specification.

Isolators (checked on every build): AMC3330/AMC3302, TPSI3050–Q1, CNY65B, contactor and relay coil–contact. Declared HV–PE crossings: MOV3 of each varistor Y, Y capacitors, the PE–side SiC of each insulation–measurement pair, the PE–end element of the AUX dividers.

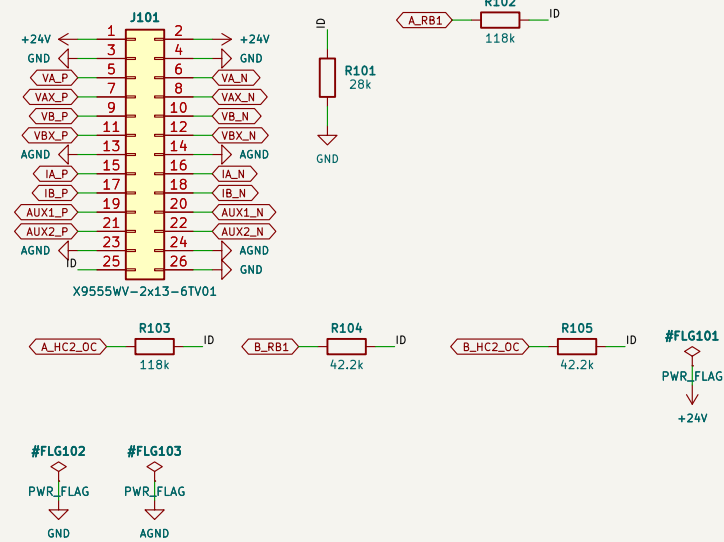
IC–17/IC–18: the whole board is conformal–coated to PD1 (IEC 60664–3 Type 1) over all HV copper and every HV–PELV isolator footprint (8–8.5 mm packages < 10 mm reinforced at PD2). Only the NH fuse links / bases and the DC terminals sit in the cooling–air path (PD3); the board, varistor network and branch fuses are in the closed PD2 compartment.

Surge credit (D–032, IC–01/02/03): the on–board varistor Y limits pole–PE and pole–pole to 3.8 kV at In 3 kA (calculated) and is monitored: reinforced is dimensioned for 6 kV (AMC3302 6.25 kV, AMC3330 7.7 kV pass on this case only). IC–04 OPEN: the G7L coil–contact rating (4000 V AC) is below reinforced – fix in rev F1 (own isolated coil supply).

SYS–IO–AUX: K_DISCH energised first at power–up; the port block itself inhibits the discharge of a closed port. Coil flyback never reaches the DO clamps in normal operation (two–switch coil drives, varistor / zener across each coil).

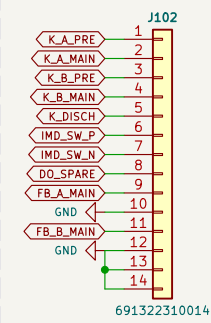
Not bench–validated. Values: sim/port_design.py → port_spec.json		
Sheet 1		
Schematics authored by Chinmoy Bhuyan		
HV–PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV–PE basic + bonding		
Sheet: /		
File: PV–PORT–180.kicad_sch		
Title: PV–PORT 180 A port board – Cover / Hierarchy		
Size: A3	Date: 2026–10–04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 1/7

PORT CONNECTOR (CTRL-C2000) + ID WITH HOLD READBACK



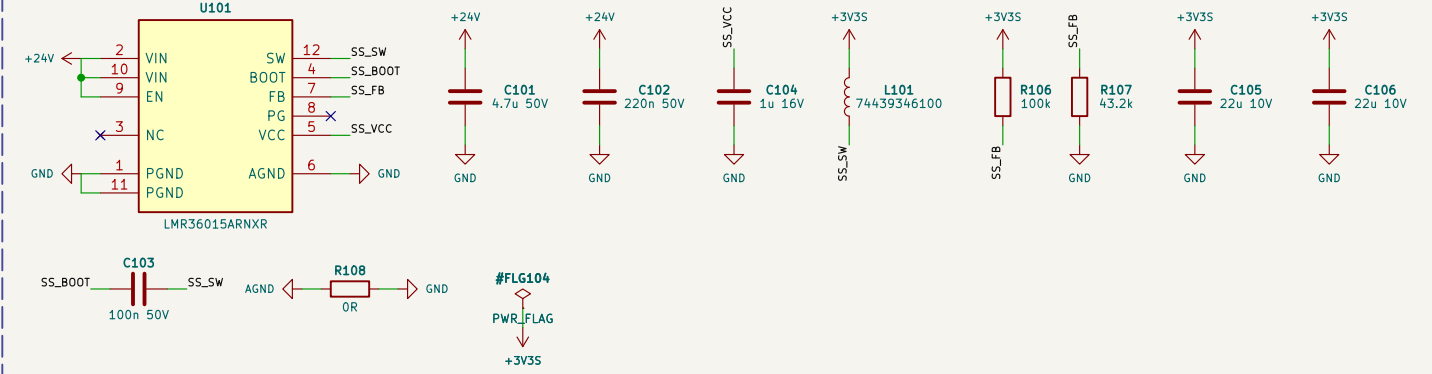
interfaces.PORT, ribbon 1:1. AMC outputs drive the cable directly (≤ 500 pF, AMC3330 p4); anti-alias RC is on CTRL-C2000. ID: 28k || readback network = 9.99 k (PV-PORT code 10.0k; CTRL-C2000 rev D values). RB outputs through 2 x 118k (port A) and 2 x 42.2k (port B): 195 states, ≥ 155 counts apart (81 with the hold coils on, sim/port_design.py sec. 7.4); CTRL decodes them with its port-only table.

CONTROL CONNECTOR (SYS-IO-AUX)



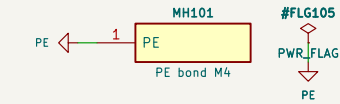
port.CTRL_PINS: DO1-DO8 (DO_SPARE = proof test), FB_A + its GND, FB_B + its GND (INT-02), 2 coil returns. Port B (battery) drives 2 HVC43 coils: K_B_MAIN up to 0.60 A -> SYS DO4 needs ILIM 10 k (1.58-2.30 A).

SENSOR SUPPLY +3V3S



LMR36015 400 kHz, 24 V -> 3.315 V (100 k / 43.2 k, SNVSB49D sec 10.2), <= 0.34 A for 6 x AMC3330 + 2 x AMC3302 + hold logic. AGND joins GND only here (0 ohm).

PE BOND



Board PE to chassis at the SPD / Y-capacitor end.

Not bench-validated. Values: `sim/port_design.py -> port_spec.json`
 Sheet 2
 Schematics authored by Chinmoy Bhuyan
 HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

Sheet: /01 Connectors and sensor supply/
File: 01_io.kicad_sch

Title: PV-PORT 180 A port board – Connectors and sensor supply

Size: A3	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 2/7

Figure 1: Schematic diagram of the TNPV12061.00M00BEEA (U301) showing its internal structure and external connections. The central component is a yellow box labeled 'U301' with pins 1 through 16. Pin 1 is VA_IN, Pin 2 is BUS, Pin 3 is VA_DCO, Pin 4 is VA_HLDO, Pin 5 is VA_HLDO, Pin 6 is INP, Pin 7 is INN, Pin 8 is DCD_C_OUT, Pin 9 is DCD_C_IN, Pin 10 is OUTP, Pin 11 is VA_P, Pin 12 is INN, Pin 13 is VDD, Pin 14 is DIAG, Pin 15 is LDO_OUT, Pin 16 is VA_LDO. External components include capacitors C302 (1n 50V), C303 (1u 25V), C304 (100n 50V), C305 (1u 25V), C306 (1n 50V), C307 (100n 50V), C308 (100n 50V), and C309 (1n 50V). The diagram shows connections to +3V3S, AGND, and various signal lines like VA_D1, VA_D2, VA_D3, VA_D4, VA_D5, VA_D6, VA_D7, VA_D8, VA_D9, VA_D10, VA_D11, VA_D12, VA_D13, VA_D14, VA_D15, VA_D16, VA_D17, VA_D18, VA_D19, VA_D20, VA_D21, VA_D22, VA_D23, VA_D24, VA_D25, VA_D26, VA_D27, VA_D28, VA_D29, VA_D30, VA_D31, VA_D32, VA_D33, VA_D34, VA_D35, VA_D36, VA_D37, VA_D38, VA_D39, VA_D40, VA_D41, VA_D42, VA_D43, VA_D44, VA_D45, VA_D46, VA_D47, VA_D48, VA_D49, VA_D50, VA_D51, VA_D52, VA_D53, VA_D54, VA_D55, VA_D56, VA_D57, VA_D58, VA_D59, VA_D60, VA_D61, VA_D62, VA_D63, VA_D64, VA_D65, VA_D66, VA_D67, VA_D68, VA_D69, VA_D70, VA_D71, VA_D72, VA_D73, VA_D74, VA_D75, VA_D76, VA_D77, VA_D78, VA_D79, VA_D80, VA_D81, VA_D82, VA_D83, VA_D84, VA_D85, VA_D86, VA_D87, VA_D88, VA_D89, VA_D90, VA_D91, VA_D92, VA_D93, VA_D94, VA_D95, VA_D96, VA_D97, VA_D98, VA_D99, VA_D100.

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Figure 8-2 is a schematic diagram of the U303 (AMC3302DWE) component. The component is represented by a yellow box with pins 1 through 16. The internal connections within the box are as follows:

- Pin 1 (IA_INP) is connected to Pin 2 (INP).
- Pin 3 (IA_N) is connected to Pin 4 (INN).
- Pin 5 (IA_P) is connected to Pin 6 (OUTP).
- Pin 7 (IA_D) is connected to Pin 8 (OUTN).
- Pin 9 (IA_D) is connected to Pin 10 (DIAG).
- Pin 11 (IA_DCO) is connected to Pin 12 (DCDC_OUT).
- Pin 13 (HLDO_IN) is connected to Pin 14 (HLDO_IN).
- Pin 15 (IA_HLDO) is connected to Pin 16 (HLDO_OUT).
- Pin 17 (IA_X) is connected to Pin 18 (NC).
- Pin 19 (IA_X) is connected to Pin 20 (DCDC_HGND).
- Pin 21 (IA_X) is connected to Pin 22 (DCDC_GND).
- Pin 23 (IA_X) is connected to Pin 24 (GND).

The external connections and components are as follows:

- Pin 1 (IA_INP) is connected to Pin 2 (INP) through a resistor R315 (10R).
- Pin 3 (IA_N) is connected to Pin 4 (INN) through a resistor R316 (10R).
- Pin 5 (IA_P) is connected to Pin 6 (OUTP) through a resistor R317 (47k).
- Pin 7 (IA_D) is connected to Pin 8 (OUTN) through a resistor R317 (47k).
- Pin 9 (IA_D) is connected to Pin 10 (DIAG) through a resistor R317 (47k).
- Pin 11 (IA_DCO) is connected to Pin 12 (DCDC_OUT) through a capacitor C320 (1n 50V).
- Pin 13 (HLDO_IN) is connected to Pin 14 (HLDO_IN) through a capacitor C321 (1u 25V).
- Pin 15 (IA_HLDO) is connected to Pin 16 (HLDO_OUT) through a capacitor C323 (1u 25V).
- Pin 17 (IA_X) is connected to Pin 18 (NC) through a capacitor C324 (1n 50V).
- Pin 19 (IA_X) is connected to Pin 20 (DCDC_HGND) through a capacitor C325 (100n 50V).
- Pin 21 (IA_X) is connected to Pin 22 (DCDC_GND) through a capacitor C326 (100n 50V).
- Pin 23 (IA_X) is connected to Pin 24 (GND) through a capacitor C327 (1n 50V).
- Pin 25 (IA_X) is connected to Pin 26 (GND) through a diode #FLG301 (PWR_FLAG).

The power and ground connections are as follows:

- Pin 1 (IA_INP) is connected to Pin 2 (INP) through a resistor R315 (10R).
- Pin 3 (IA_N) is connected to Pin 4 (INN) through a resistor R316 (10R).
- Pin 5 (IA_P) is connected to Pin 6 (OUTP) through a resistor R317 (47k).
- Pin 7 (IA_D) is connected to Pin 8 (OUTN) through a resistor R317 (47k).
- Pin 9 (IA_D) is connected to Pin 10 (DIAG) through a resistor R317 (47k).
- Pin 11 (IA_DCO) is connected to Pin 12 (DCDC_OUT) through a capacitor C320 (1n 50V).
- Pin 13 (HLDO_IN) is connected to Pin 14 (HLDO_IN) through a capacitor C321 (1u 25V).
- Pin 15 (IA_HLDO) is connected to Pin 16 (HLDO_OUT) through a capacitor C323 (1u 25V).
- Pin 17 (IA_X) is connected to Pin 18 (NC) through a capacitor C324 (1n 50V).
- Pin 19 (IA_X) is connected to Pin 20 (DCDC_HGND) through a capacitor C325 (100n 50V).
- Pin 21 (IA_X) is connected to Pin 22 (DCDC_GND) through a capacitor C326 (100n 50V).
- Pin 23 (IA_X) is connected to Pin 24 (GND) through a capacitor C327 (1n 50V).
- Pin 25 (IA_X) is connected to Pin 26 (GND) through a diode #FLG301 (PWR_FLAG).

$|I_A| > 410 \text{ A}$ (band 382–441 A, own REF3030E $\pm 0.1\%$ network, CM-cancelling) AND FB = closed AND
 DAD = valid $\rightarrow$ HOLD1. Test injection 86.6 μ from A_TST_3 (side A) / A_TST_TD (side B): shifts the threshold
 by 459 A, so a healthy side trips at 0 A under TEST only.

$I_{IA} > 410$ A (band 392-441 A, own REF3030E + 0.1 % network CM-cancelling) AND FB = closed AND
 DAG = valid $\rightarrow$ HCLD2: Test Injection 86.6 μ A from A_TST_T2A (side A) / A_TST_T2B (side B): shifts the threshold
 by 459 A, so a healthy side trips at 0 A under TEST only.

100 k / 15 k $\rightarrow$ Schmitt buffer T (ch1 side A). T $\rightarrow$ 47 k / 22 uF $\rightarrow$ T_d (ch1 side B). One-shots (2.3 uF / 47 k) of T $\rightarrow$ T2A (ch2 side A) and of T_d $\rightarrow$ T2B (ch2 side B). No single fault trips both channels for longer than one one-shot. Firmware runs it at every start, contactor open and closed (report sec. 7.4; windows, what it finds and what it does not).

[illegible]

Not bench-validated. Values: `sim/port_design.py -> port_spec.json`
Sheet 4
Schematics authored by Chinmoy Bhuyan
HV-PELV reinforced (1000 VDC, OVC II, PD2, coated to PD1); HV-PE basic + bonding

Sheet: /03 Port A sensing and hold logic/
File: 03_portA_sense.kicad_sch

Title: *PV-PORT 180 A port board – Port A sensing and hold logic*

Size: A2	Date: 2026-10-04	Rev: F0
KiCad E.D.A. 10.0.6		Id: 4/7

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100 k / 15 k -> Schmitt buffer T (ch1 side A). T -> 47 k / 22 uF -> T_d (ch1 side B). One-shots (2.3 uF / 47 k) of T -> T2A (ch2 side A) and of T_d -> T2B (ch2 side B). No single fault trips both channels for longer than one one-shot. Firmware runs it at every start, contactor open and closed (report sec. 7.4; windows, what it finds and what it does not).

[illegible]

Not bench-validated. Values: sim/port_design.py -> port_spec.json

Sheet 6

Schematics authored by Chinmoy Bhuyan

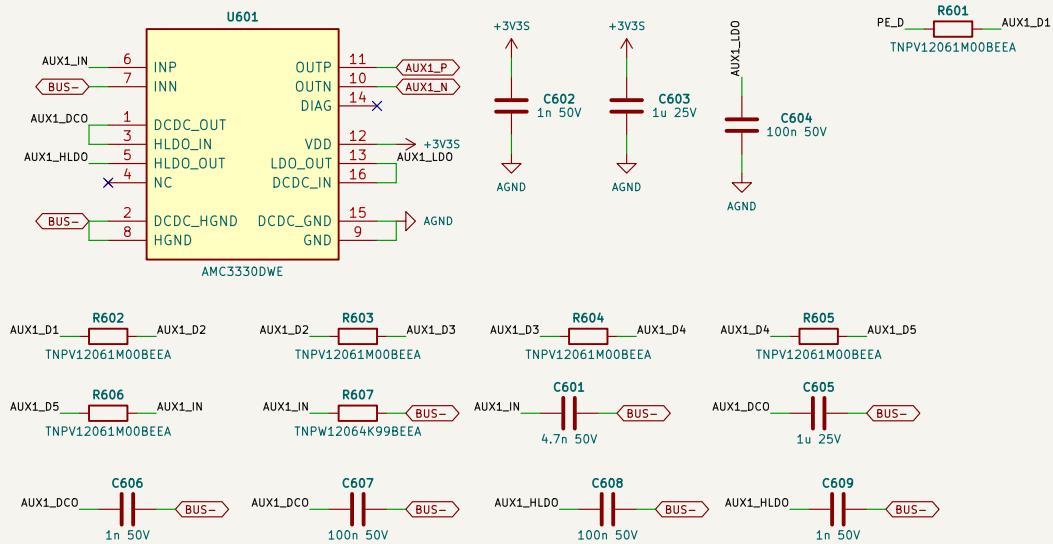
HV-PEM reinforced (1000 VDC, OVC II, PD2, coated to PD1): HV-PE basic + bonding

Sheet: /05 Port B sensing and hold logic/
File: 05_portB_sense.kicad_sch

Title: PV-PORT 180 A port board – Port B sensing and hold logic

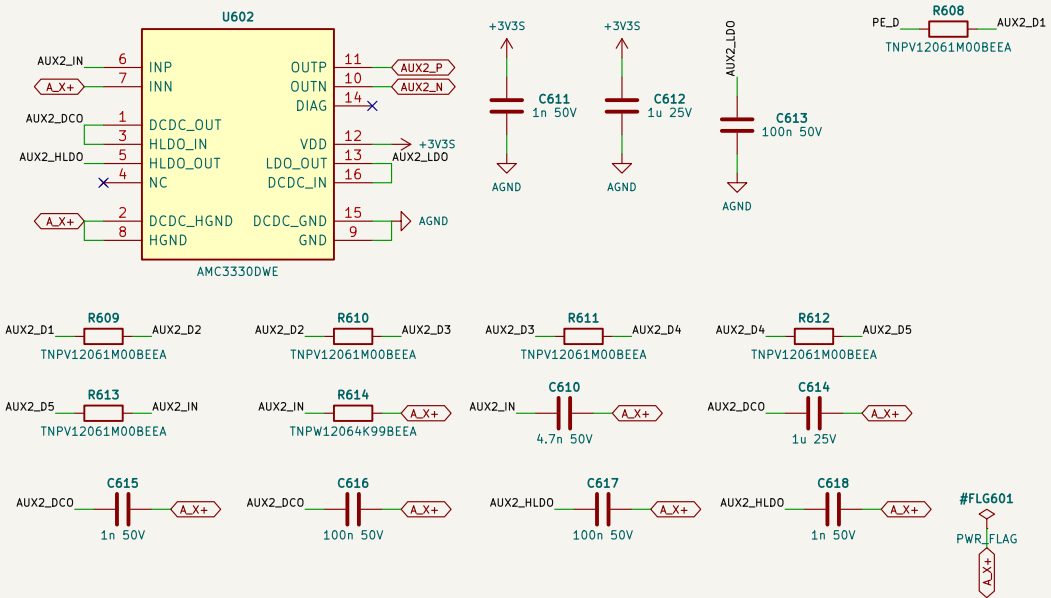
Size: A2	Date: 2026-10-04	Rev: F0
KiCad E.A. 10.0.6		Id: 6/7

AUX1 – PE RELATIVE TO BUS–



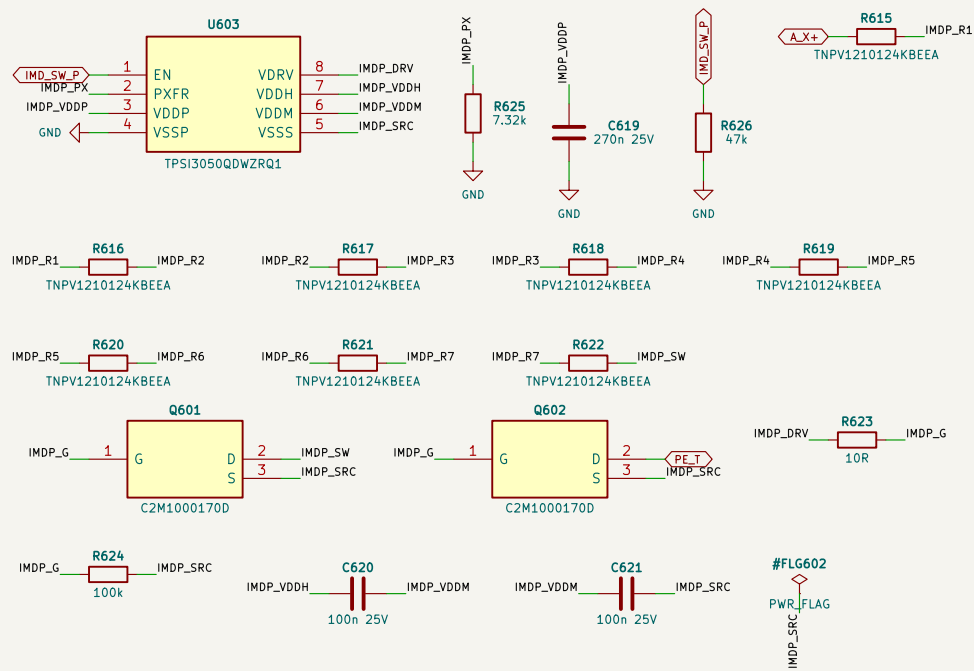
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

AUX2 – PE RELATIVE TO A_X+



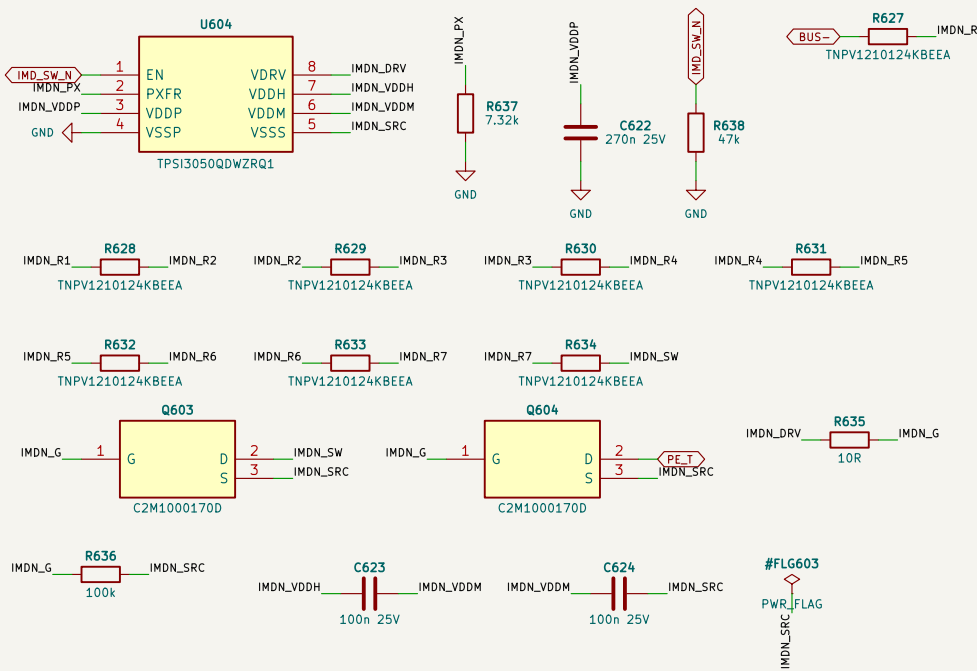
6 x 1.00 M from PE + 4.99 k: +/-1203 V linear. The divider is a fixed, known 6.005 M path to PE (included in the IMD equations, sim/port_design.py sec. 8). First element = HV–PE crossing.

TEST STRING A_X+ -> PE (IMD_SW_P)



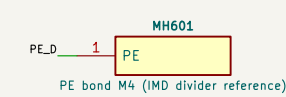
8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

TEST STRING BUS– -> PE (IMD_SW_N)



8 x TNPV1210 124 k = 992 k (125 V, 0.13 W each at 1000 V): 1 mA test current. Energised = connected. Common-source C2M1000170D pair (PA–09): blocks either polarity, so a pole below PE cannot leak through a body diode. TPSI3050 two-wire from the 24 V DO, reinforced. IDSS <= 100 uA each – EOL baseline cal.

PE_D BOND + ONE ACTIVE MONITOR PER DC SYSTEM



PE_D: own M4 bond (dividers only). The module disables its own measurement by leaving IMD_SW_P/N de-energised (strings open, fail-safe): a passive module still loads each pole with 6.005 M to PE.